

Proposal in Response to RFQ – Solar-Powered Irrigation Systems

Submitted to: Supremo Green Revolution (SGR)

Submitted by: GreenTech Irrigation Solutions Ltd. (GTIS)

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1. Executive Summary

GreenTech Irrigation Solutions Ltd. (GTIS) is proud to submit this proposal in response to the RFQ issued by Supremo Green Revolution (SGR) for the design, supply, and installation of solar-powered irrigation systems across Kenya, Ghana, and Tanzania.

GTIS is an ISO-certified engineering company headquartered in Nairobi with regional offices in Accra and Dar es Salaam. With over eight years of field experience in renewable energy and water management, we have successfully implemented 100+ solar-powered projects across East and West Africa. GTIS combines strong local technical capacity with global best practices to deliver affordable, durable, and smart irrigation systems that empower smallholder farmers.

Our proposal provides a technically sound, cost-efficient, and scalable approach that fully aligns with SGR's goals under the Green Resilience Initiative.

2. Company Profile and Capability

Founded in 2016, GreenTech Irrigation Solutions specializes in sustainable irrigation, solar engineering, and agricultural infrastructure. The company is fully certified under:

- ISO 9001:2015 – Quality Management Systems
- ISO 14001:2015 – Environmental Management Systems
- ISO 45001:2018 – Occupational Health and Safety
- IEC 62257 Compliance – Renewable Energy System Standards

GTIS employs over 85 full-time staff, including civil engineers, agronomists, and renewable energy technicians. The company maintains in-house design, logistics, and maintenance divisions, ensuring end-to-end delivery capability.

Key clients include the Kenyan Ministry of Agriculture, the Ghana Irrigation Development Authority, and the Tanzania Renewable Energy Association. GTIS has also partnered with FAO and UNDP on pilot irrigation schemes targeting smallholder communities.

3. Technical Approach

GTIS proposes a high-efficiency solar-powered irrigation solution designed for reliability, modular scalability, and ease of maintenance. Each system will provide an average daily water output of 10,000–15,000 liters, supporting between 10 and 20 farmers per site.

****System Components:****

- Solar PV arrays (3–5 kW) with Tier 1 panels and corrosion-resistant aluminum frames.
- DC submersible pump (up to 15 m head) with automatic dry-run protection.
- 5,000–10,000 L storage tanks with overflow control.

- Drip irrigation kits customized for maize, vegetables, and legumes.
- GSM-enabled smart controller for performance tracking via mobile dashboard.

****Design and Installation Workflow:****

1. Site Assessment & Engineering Design – solar and hydro surveys, electrical load analysis.
2. Procurement & Pre-assembly – component testing at GTIS Nairobi workshop.
3. Installation & Commissioning – on-site construction, wiring, and quality testing.
4. Training & Handover – three-day practical sessions for farmers and local technicians.
5. Monitoring & Support – continuous data collection and remote troubleshooting.

Each system will be fully documented, GPS-tagged, and integrated into SGR's central performance database.

4. Deliverables

1. 12 complete solar irrigation installations (4 per country).
2. Real-time monitoring dashboard accessible to SGR project managers.
3. Training for 500 farmers and 25 technicians.
4. Bilingual operation manuals (English/Swahili/Twi).
5. Two-year preventive maintenance and after-sales service plan.
6. Quarterly performance and impact reports.

5. Quality Assurance and Compliance

GTIS follows a rigorous Quality Management Plan (QMP) covering all project stages from procurement to post-installation verification. Each component supplier is vetted for compliance with ISO 9001 and IEC 61215 standards. Internal audits are conducted quarterly to ensure conformance.

Environmental and safety management practices comply with ISO 14001 and ISO 45001, including safe chemical handling, waste management, and worker safety during installation. All GTIS technicians are certified under the Kenya National Industrial Training Authority (NITA).

Our commitment to quality ensures every delivered system meets international durability and performance benchmarks.

6. Implementation Plan

Phase	Activity	Duration
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1	Feasibility Study & Site Selection	3 weeks
2	Design, Procurement & Shipping	5 weeks
3	Installation & Commissioning	6 weeks
4	Training & Handover	2 weeks
5	Monitoring & Reporting	Ongoing

Total implementation period: ****16 weeks**** (approx. four months).

GTIS will deploy local project leads in each country and maintain regional spare-part depots for rapid servicing. A web-based monitoring dashboard will be provided to SGR's technical team to review real-time performance indicators.

7. Financial Proposal (Indicative, USD)

Item	Quantity	Unit Cost (USD)	Subtotal (USD)
Solar Irrigation System (complete package)	12	7,200	86,400
Farmer Training & Capacity Building	12	2,500	30,000
Monitoring & Reporting System	1	10,000	10,000
Spare Parts & Maintenance Kits	12	600	7,200
Logistics, Insurance, and Taxes	1	14,000	14,000
Total Estimated Cost (USD)			**147,600**

- Notes:
- Inclusive of two-year service contract.
 - Costs exclude import duty variations by country.
 - Valid for 90 days from submission.
 - Payment schedule: 30% advance, 50% upon installation, 20% after acceptance.

8. Sustainability and After-Sales Service

GTIS guarantees two years of full warranty and technical support for all installations. We will train and certify local “Irrigation Champions” in each region to provide front-line maintenance and troubleshooting.

- Our sustainability framework includes:
- Remote diagnostics and predictive maintenance alerts.
 - Stocked spare-parts depots in Nairobi and Accra.
 - Annual community workshops on cooperative management and financial sustainability.
 - Optional expansion packages allowing systems to double capacity with minimal reconfiguration.

9. Strengths and Risks

- **Strengths:****
- Complete ISO certification across quality, environment, and safety.
 - Local presence and logistics capacity in all three target countries.
 - Smart monitoring and data-driven maintenance.
 - Proven track record with government and donor-funded irrigation programs.

- **Risks / Limitations:****
- Slightly higher unit cost due to advanced components.
 - Implementation schedule dependent on customs clearance timelines.

10. Conclusion

GreenTech Irrigation Solutions Ltd. offers a robust, fully compliant, and technically advanced solution for SGR’s solar irrigation program. Our systems are engineered for reliability, designed for local conditions, and supported by a dedicated service network across Africa.

GTIS is committed to delivering beyond compliance — combining innovation, data transparency, and long-term community empowerment. We believe this partnership with Supremo Green Revolution (SGR) can set a benchmark for renewable irrigation in Africa.

Submitted by:

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